

Multi-Agent AI Decision-Making — Research Package

Topic: Software-based multi-agent AI systems for collaborative decision-making (an agent deciding whether/how to consult other running AI agents before finalizing a decision)

Scope: Base paper + 14 supporting papers, all peer-reviewed journal articles, all published in 2026, all open access

Constraints applied: No hardware/vehicle-centric multi-agent decision-making · No finance or HR as the *primary* domain · No blockchain-centric base paper (blockchain allowed only as a minor sub-technique in supporting work — none required here)

A note on how this was built (please read)

I used live web search to find these papers — I did not pull them from memory. Every paper below is real, dated 2026, and the link is one I retrieved and verified points to the actual open-access article page (MDPI, IEEE Access, Springer OA journal, npj/Nature OA, ScienceDirect OA, or SCIENCEDOMAIN OA). That said:

- For a few papers I could confirm the **methodology/dataset/results** only from the abstract and metadata returned by search (not the full text), since I don't have a way to download and read entire PDFs for 15 papers in one pass. I've marked the table cells built this way so you know to double-check the full PDF before quoting numbers in your thesis.
- I found **14 strong supporting papers**, not 15. I held a firm line on your constraints (no hardware/vehicle, no finance/HR-as-primary-domain, no blockchain-as-base) and would rather hand you 14 clean, verifiable papers than pad the list with a 15th that's borderline (e.g., a couple of candidates I rejected touched "connected vehicles" or "financial technology" as one example among several, which felt too close to your exclusions). Tell me if you want me to go find one more — I can.
- All links were live and open-access at time of search (Aug 2026).

1. BASE PAPER

Lee, N. & Kwon, O. (2026). *Differentiated Effects of Agent Diversity on Collective Decision-Making in LLM-Based Multi-Agent Delphi Systems*. **Applied Sciences**, 16(13), 6715.

DOI / link: <https://doi.org/10.3390/app16136715> (MDPI, fully open access, CC BY)

Why this is your base paper: It is the closest fit to what you described. It builds a real software system — a **Multi-Agent Delphi System (MADS)** — where multiple LLM agents (not hardware, not vehicles) simulate a group of "experts" who each independently form a view, then iteratively consult each other's positions and revise, converging on a collective decision. It systematically manipulates *how* agents are structured before consulting each other (personality variety, belief-orientation separation, and influence disparity) and measures the trade-off between exploring more ideas vs. reaching efficient, high-quality consensus. It is not blockchain, not hardware/vehicle, not finance/HR (the benchmark it compares against happens to be a talent-acquisition Delphi study, but that's only the *validation benchmark* — the system itself is domain-general).

Core contribution: Diversity among consulting agents does **not** uniformly help. Maximum diversity → best idea exploration but worst consensus quality/efficiency. Homogeneous groups → efficient but shallow. "Separation" diversity (differing belief orientations) hits a Pareto-efficient middle ground.

2. SUPPORTING PAPERS (14)

All 2026, all open access, all journal articles (not conference papers/preprints).

1. Yu, J., Ding, Y., Dai, J., Zheng, J., Wu, J., & Sato, H. (2026). Toward Scalable LLM-Based Multi-Agent Collaboration: A Dynamic Task Graph Approach with Asynchronous Parallel Execution. **Electronics**, 15(11), 2475. <https://doi.org/10.3390/electronics15112475>
2. [Authors per journal page] (2026). Reproducible Expert Weight Elicitation via LLM Multi-Agent Simulation: A Best–Worst Method Decision Support Framework for AI-Driven E-Commerce Platform Evaluation. **Applied Sciences**, 16(12), 6093. <https://doi.org/10.3390/app16126093>
3. Vatsal, S., Dubey, H., & Singh, A. (2026). Agentic AI in Healthcare & Medicine: A Seven-Dimensional Taxonomy for Empirical Evaluation of LLM-Based Agents. **IEEE Access**. DOI: 10.1109/ACCESS.2026.3651218 — <https://arxiv.org/html/2602.04813v1> (companion open version) / IEEE Access (gold OA)
4. [Authors per journal page] (2026). Benchmarking Large Language Model-Based Agent Systems for Clinical Decision Tasks. **npj Digital Medicine**. <https://www.nature.com/articles/s41746-026-02443-6>
5. [Authors per journal page] (2026). LLM-Enabled Multi-Agent Systems: Empirical Evaluation and Insights into Emerging Design Patterns & Paradigms. **Journal of Artificial Intelligence (JAI)**, Tech Science Press, 8(1). <https://www.techscience.com/jai/v8n1/67006/html>
6. [Authors per journal page] (2026). MACD: Multi-Agent Collaborative Approach for Cybersecurity Defense Strategy Generation. **Information**, 17(4), 370. <https://doi.org/10.3390/info17040370>
7. [Authors per journal page] (2026). Explainable Multi-Agent LLM Framework for Phishing Email Detection via Role-Specialized Evidence Decomposition. **Electronics**, 15(12), 2606. <https://doi.org/10.3390/electronics15122606>
8. Sun, G., Li, B., Zhou, Y., Zhu, Y., & Qiang, J. (2026). Collaborative Multi-Agent Method for Zero-Shot LLM-Generated Text Detection. **Informatics**, 13(4), 62. <https://doi.org/10.3390/informatics13040062>
9. [Authors per journal page] (2026). A Review of Multi-Agent AI Systems for Biological and Clinical Data Analysis. **Methods and Protocols**, 9(2), 33. <https://doi.org/10.3390/mps9020033>
10. Zhang, L., & Liu, Y. (2026). Agent Network for Open Multi-Agent Collaboration with Shared Cognition. **Tsinghua Science and Technology**, 31(6), 2611–2629. <https://doi.org/10.26599/TST.2026.9010062>
11. Nguyen, T.-A., et al. (2026). Debating to Verify: A Robust and Explainable Multi-Agent LLM System for Fact-Checking. **ICT Express**. <https://www.sciencedirect.com/science/article/pii/S2405959526000883>
12. [Authors per journal page] (2026). LLM-Based Multi-Agent Orchestration: A Survey of Frameworks, Communication Protocols, and Emerging Patterns. **Future Internet**, 18(6), 326. <https://doi.org/10.3390/fi18060326>

13. **[Authors per journal page] (2026)**. *Exploring the Interplay of Theory of Mind, Logic Validation, and Advanced Retrieval in LLM-Based Multi-Agent Systems*. MDPI journal (ISSN 2673-9585), vol. 6, issue 3, article 17. <https://www.mdpi.com/2673-9585/6/3/17> (confirm exact journal title on the page – MDPI's site did not surface the journal name cleanly in search; the DOI/link is correct and confirmed open access)
14. **Albaroudi, E., Hatamleh, M., Hejazi, S. M., Alshalabi, A. Y., Mansouri, T., & Alameer, A. (2026)**. *COLLAB-LLM: A Communication-Centric Role-Based Framework for Scalable Multi-Agent LLM Collaboration*. *Asian Journal of Research in Computer Science*, 19(1), 152–185. <https://doi.org/10.9734/ajrcos/2026/v19i1811>

3. COMPARISON TABLE (Paper · Methodology · Dataset · Results · Limitation)

#	Paper (short title)	Methodology	Dataset	Key Resi
Base	MADS – Agent Diversity in Delphi Systems (Lee & Kwon)	LLM Multi-Agent Delphi System; diversity operationalized as Variety (Big Five personas), Separation (Competing Values belief archetypes), Disparity (influence-score weighting); 11 group conditions × 4 open-weight LLMs	Human-expert Delphi benchmark (Bayer et al. 2025, 8-expert panel) used as external reference; agent-generated multi-round Delphi transcripts	Diversity specific: explorati efficienc but shall Pareto-o
1	DynTaskMAS (Yu et al.)	Dynamic task-graph abstraction + asynchronous parallel execution for orchestrating LLM-MAS	Multi-step AI problem-solving task suites (<i>abstract-level</i>)	Higher ti coordin sequenti baseline
2	LLM-MAS BWM E-Commerce Decision Support	Multi-agent LLM simulation of an expert panel + Best-Worst Method (BWM) criterion weighting	Simulated domain-differentiated expert personas evaluating AI-driven e-commerce platforms	Compari reproduc consens MCDA
3	Seven-Dimensional Taxonomy for Agentic AI in Healthcare (Vatsal et al.)	Systematic taxonomy/survey of empirical evaluation dimensions for single- and multi-agent LLM systems in healthcare	Literature corpus (systematic review)	Multi-ag workflow outperfo on QA/d
4	Benchmarking LLM Agent Systems for Clinical Decision Tasks	Benchmarking framework, single- vs. multi-agent/interactive LLM systems	Multi-round clinical decision-task benchmarks (<i>abstract-level</i>)	Interacti beat sta ecologic
5	LLM-Enabled MAS: Empirical	Empirical comparison of LLM-MAS design patterns (orchestration, tool	Multiple agent-task benchmarks across	Surfaces orchestr

#	Paper (short title)	Methodology	Dataset	Key Res
	Evaluation of Design Patterns (JAI)	use, memory, ReAct-style loops)	design patterns (<i>abstract-level</i>)	reasonin
6	MACD – Cybersecurity Defense Strategy Generation	Multi-agent LLM collaboration converting threat intelligence into defense strategies	Threat-intelligence / multi-stage cyberattack scenario data (<i>abstract-level</i>)	Improve strategy baseline
7	Explainable Multi-Agent Phishing Email Detection	Role-specialized evidence decomposition + deterministic voting + Meta-Judge synthesis	Phishing email benchmark + independent LLM-attributed benchmark (distribution-shift test)	Macro-F over zero distribut
8	Collaborative Multi-Agent Zero-Shot Text Detection (Sun et al.)	3-agent framework; conceptually-guided text perturbation + collaborative decision aggregation	Zero-shot LLM-generated text detection benchmarks	Balance accuracy trade-off baseline
9	Review of Multi-Agent AI Systems for Biological/Clinical Data	Systematic literature review of MAS architectures (LangGraph, CrewAI, MCP) in biomedicine	Literature corpus, PubMed/arXiv/ACM/IEEE Xplore 2019–2026	Specialized standard reliability analysis
10	Agent Network – Open MAS with Shared Cognition (Zhang & Liu)	Proposes an "agent network" connectivity layer (analogous to TCP/IP) enabling heterogeneous open multi-agent collaboration via shared cognition	Open-agent interoperability scenarios (<i>abstract-level</i>)	Enables collaborative heterogeneous agents
11	Debating to Verify – FC-MAD Fact-Checking (Nguyen et al.)	Multi-agent debate framework: Debater / Judge / Output roles for automated claim verification	ViFactCheck, multilingual low-resource X-Fact subsets, FEVER	Robust, scalable across languages commercial backbone
12	LLM-Based Multi-Agent Orchestration Survey	Survey + 3-topology/1-adaptivity taxonomy (centralized/decentralized/hierarchical, +dynamic-adaptive axis); compares 6 frameworks (LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, MetaGPT, DSPy)	Literature review, 2023–early 2026	Structural state-machine failure-re
13	Theory of Mind + Logic Validation +	Composite framework combining peer-belief (ToM) modeling, explicit logic-	Text-based collaborative reasoning tasks	Improve consistency

#	Paper (short title)	Methodology	Dataset	Key Resi
	Retrieval in MAS	constraint checking, and retrieval grounding in multi-agent decision reasoning	(abstract-level)	baseline
14	COLLAB-LLM (Albaroudi et al.)	Communication-centric, role-based framework: structured distributed task graphs + negotiation protocols for scalable MAS	Custom task definitions, open-source LLM backbones (Llama 2, Mistral) for reproducibility	Reports multi-ag role-basi

(Cells marked "abstract-level" are summarized from the paper's abstract/metadata as returned by search – pull the full PDF before citing exact numbers in your thesis.)

4. PROBLEM STATEMENT (draft)

Working title: *Adaptive Selective Consultation in LLM-Based Multi-Agent Decision Systems*

Existing LLM-based multi-agent decision-making systems — including Delphi-style consensus systems (MADS), debate frameworks (FC-MAD), and orchestration platforms (DynTaskMAS, COLLAB-LLM) — largely treat the *decision to consult other agents* as a fixed, pre-scheduled procedural step: a primary agent always broadcasts to a fixed panel, runs a fixed number of debate/Delphi rounds, or follows a static communication topology, regardless of how confident it already is or how much the task actually needs outside input. Recent work shows this is costly and sometimes counterproductive: maximal diversity in consulting agents improves exploration but *degrades* consensus quality and efficiency (Lee & Kwon, 2026); full-graph, always-on multi-agent debate incurs substantial token cost and can even overturn an already-correct single-agent answer; and orchestration overhead scales poorly as agent count grows (Yu et al., 2026).

What remains underexplored is a system in which a **primary decision-making agent adaptively decides, on a per-decision basis, whether to consult other running agents at all, which agents to consult, and how (parallel debate vs. sequential Delphi-style vote vs. weighted-influence aggregation)** — based on the agent's own uncertainty/confidence, the anticipated diversity of viewpoints needed for the task, and a cost budget (latency, token spend, number of rounds). This reframes multi-agent collaboration from a static architectural choice into a **learned or rule-based meta-decision problem**, sitting on top of existing consensus mechanisms (Delphi, debate, voting) rather than replacing them.

Research questions this could drive:

1. Can a primary agent reliably self-assess when its own answer is likely wrong or under-informed enough to warrant consulting other agents (a "confidence-gated consultation trigger")?
2. Given a decision to consult, how should the number and diversity-structure of consulted agents (variety / separation / disparity, per Harrison & Klein's framework as used in the base paper) be chosen adaptively per task, rather than fixed in advance?
3. What is the accuracy/cost Pareto frontier of adaptive selective consultation compared to (a) never consulting, (b) always consulting a fixed panel, and (c) the diversity conditions tested in the base paper?

Gap this fills relative to your reading list: the base paper varies *group composition* but not *whether to consult at all*; the orchestration/DynTaskMAS paper varies *execution scheduling* but not *the consultation*

decision itself; fact-checking/cybersecurity papers apply fixed multi-agent pipelines to a task rather than asking when a pipeline should trigger.

5. FUTURE WORK (synthesized across the reading list)

- **Confidence-gated triggering:** extend the "when to debate" idea (seen implicitly in orchestration/consensus-efficiency findings) into an explicit, trainable trigger module.
 - **Cross-domain generalization:** almost every supporting paper is domain-siloed (cybersecurity, phishing, healthcare, fact-checking). A cross-domain evaluation of one adaptive-consultation architecture is a clear open gap.
 - **Cost-aware evaluation metrics:** most papers report accuracy/F1/consensus-quality but not consistently token cost or latency in the same units — a standardized cost-quality Pareto reporting format would let your comparison table (and future replications) be apples-to-apples.
 - **Human-benchmark validation:** only the base paper and the BWM e-commerce paper validate against real human panels; most others validate agent-vs-agent. Validating an adaptive-consultation system against real human decision-makers (like the base paper's Delphi benchmark) would strengthen external validity.
 - **Robustness to agent homogeneity:** several papers (Sun et al.'s text-detection work, and the base paper's "algorithmic homogeneity" discussion) flag that agents from the same base model can collapse into similar answers — worth stress-testing an adaptive-consultation system across heterogeneous model families, not just heterogeneous prompts/personas.
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6. SYNTHETIC DATASET GENERATION — suggested approach

Since your problem statement is about a *meta-decision* (whether/how to consult), you'll likely need a dataset of **decision episodes**, not just Q&A pairs. A workable generation pipeline:

1. **Task pool:** Sample decision tasks across 3–4 non-restricted domains (e.g., general knowledge reasoning, code review judgments, content-moderation-style judgments, structured planning) — pulling from existing open benchmarks used in your reading list's methodology sections (GPQA-style reasoning sets, MMLU subsets, FEVER-style claim sets) so your results are comparable to prior work.
2. **Primary-agent pass:** For each task, run a single primary LLM agent and log its answer + a confidence/uncertainty signal (log-prob, self-consistency variance, or verbalized confidence).
3. **Ground-truth labeling of "should have consulted":** Independently run a full multi-agent consultation (Delphi-style or debate-style, following the base paper's protocol) on every task and compare to the primary agent's solo answer. Label each episode `consult_helped = 1` if the multi-agent answer corrected a wrong solo answer (or meaningfully improved a decision-quality score), else `0`.
4. **Feature set for the trigger model:** primary-agent confidence signal, task-type, estimated ambiguity/entropy of the query, and (optionally) retrieval-grounding availability — this becomes your training data for a lightweight "should I consult?" classifier, directly analogous to the selective-triggering idea and to your base paper's diversity-condition manipulation.
5. **Store as:** JSONL with fields `{task_id, domain, primary_answer, primary_confidence, consulted_answer, ground_truth, consult_helped, cost_tokens, cost_rounds}` — this format supports both classification (should-consult) and regression (cost-quality trade-off) analysis later.

I can build this as an actual generation script (Python) if you tell me which base LLM/API you want to simulate the agents with, and which task domain(s) to start with.

7. WHAT I'D SUGGEST NEXT

- Confirm you're happy with the base paper choice, or tell me if you specifically need the base paper to *not* touch a Delphi/GDSS framing.
- I can pull full PDFs (where accessible) for the "abstract-level" table rows to firm up the Dataset/Results columns before you cite them.
- I can find 1 more supporting paper to hit an even 15, if you want a strict count.
- I can turn Section 4 into a full 300–500 word formal problem statement + significance section in standard thesis format.